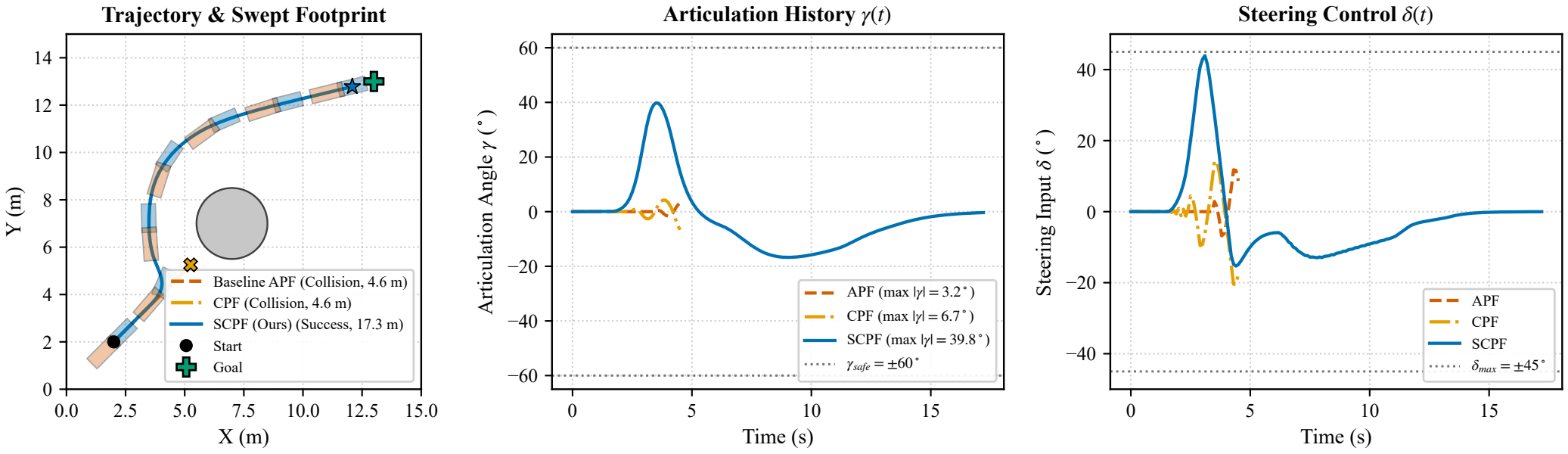

Trajectory & Swept Footprint
X (m)
Y (m)
Baseline APF (Collision, 4.6 m)
CPF (Collision, 4.6 m)
SCPF (Ours) (Success, 17.3 m)
Start
Goal
Articulation History $\gamma(t)$
Articulation Angle γ (°)
Time (s)
APF (max $|\gamma| = 3.2°$)
CPF (max $|\gamma| = 6.7°$)
SCPF (max $|\gamma| = 39.8°$)
$\gamma_{safe} = \pm 60°$
Steering Control $\delta(t)$
Steering Input δ (°)
Time (s)
APF
CPF
SCPF
$\delta_{max} = \pm 45°$